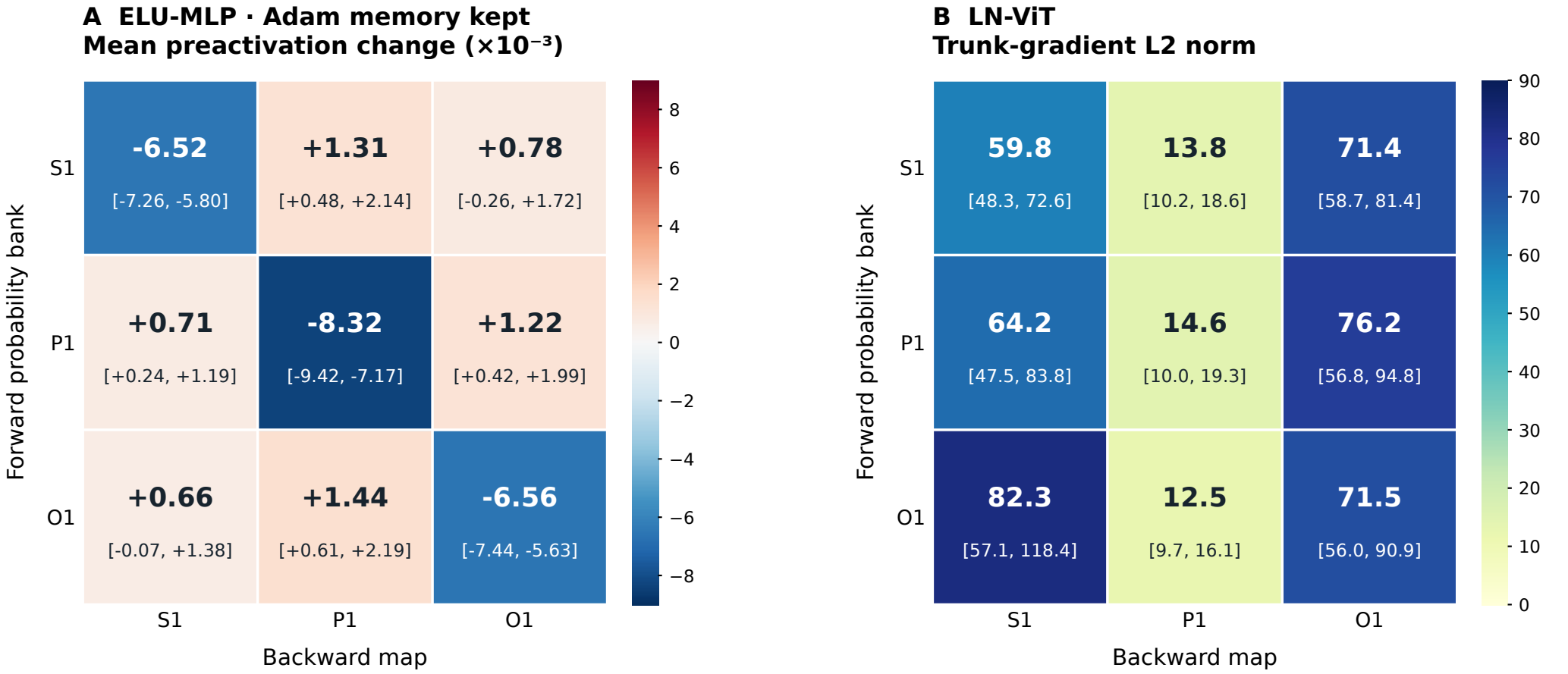
Crossed forward probabilities and backward maps

Actual new labels · native Jacobian · 10 ELU-MLP seeds / 5 LN-ViT seeds



Each row reuses every example's complete probability vector within each seed. Across columns, only the backward map changes.

Cells: existing seed mean [unadjusted descriptive 95% seed-bootstrap CI]. Cell intervals describe means, not between-cell differences.

S: learned map. P: independent class-row permutations by column (changes W^TW). O: class-space orthogonal map (preserves W^TW).

S1, P1 and O1 have equal centered head Frobenius norm within each source; this does not imply identical residual-to-trunk maps.

ELU: negative means lower full-set recipient preactivation after one virtual Adam step. Gradients use the fixed 128-example subset.

ViT: mean-loss trunk gradient on those 128 examples. These frozen quantities do not assign causal shares of later learning delay.